

Vector Fields and Line Integrals

ANSWER KEY



Vector fields and divergence

- 1 For the vector field $\mathbf{F}(x, y) = \langle x^2y, xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.
 $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(xy^2) = 2xy + 2xy = 4xy.$
- 2 For $\mathbf{F}(x, y, z) = \langle y, -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.
 $\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(z^2) - \frac{\partial}{\partial z}(-x), \frac{\partial}{\partial z}(y) - \frac{\partial}{\partial x}(z^2), \frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y) \right\rangle = \langle 0, 0, -2 \rangle.$

Line integrals

- 3 Evaluate the line integral $\int_C (x + y) ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.
Parametrize $x = 3t, y = 4t, 0 \leq t \leq 1$: $ds = 5 dt$. $\int_0^1 (3t + 4t)(5) dt = 35 \int_0^1 t dt = \frac{35}{2}.$
- 4 Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle, 0 \leq t \leq 1$.
 $\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t \rangle, \mathbf{r}'(t) = \langle 1, 2t \rangle$: $\int_0^1 (t^2 + 2t^2) dt = \int_0^1 3t^2 dt = 1.$

Conservative vector fields and potential functions

- 5 Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.
 $P = 2xy$: $\frac{\partial P}{\partial y} = 2x$. $Q = x^2 + 3y^2$: $\frac{\partial Q}{\partial x} = 2x$. Since these are equal, $\mathbf{F}$ is conservative.
- 6 Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.
 $f_x = 2xy \Rightarrow f = x^2y + g(y)$. Then $f_y = x^2 + g'(y) = x^2 + 3y^2$, so $g'(y) = 3y^2$, giving $g(y) = y^3$. Potential function: $f(x, y) = x^2y + y^3$ (plus any constant).

The Fundamental Theorem for line integrals

- 7 Using the potential function $f(x, y) = x^2y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.
 $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = (1 \cdot 2 + 8) - (0) = 10.$

Green's Theorem

- 8 Use Green's Theorem, $\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$, to evaluate $\oint_C xy dx + x^2 dy$ where C is the boundary of the square $0 \leq x \leq 1, 0 \leq y \leq 1$.
 $P = xy, Q = x^2$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - x = x$. $\iint_D x dA = \int_0^1 \int_0^1 x dy dx = \int_0^1 x dx = \frac{1}{2}.$

- 9 Use Green's Theorem to find the area of the region enclosed by a curve C , using the formula $A = \frac{1}{2} \oint_C (x dy - y dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \leq t \leq 2\pi$.
 $dx = -2\sin t dt$, $dy = 2\cos t dt$.
 $x dy - y dx = 2\cos t(2\cos t dt) - 2\sin t(-2\sin t dt) = 4\cos^2 t dt + 4\sin^2 t dt = 4 dt$.
 $A = \frac{1}{2} \int_0^{2\pi} 4 dt = 4\pi$ (matching the known area of a radius-2 circle).
- 10 Use Green's Theorem to evaluate $\oint_C (3y) dx + (5x) dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.
 $P = 3y$, $Q = 5x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 5 - 3 = 2$. $\iint_D 2 dA = 2 \cdot 4\pi = 8\pi$, using that the disk of radius 2 has area 4π .

Work done by a force field

- 11 A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.
 Leg 1 ($y = 0$, $x: 0 \rightarrow 2$): $\mathbf{F} = \langle 0, x \rangle$, $d\mathbf{r} = \langle dx, 0 \rangle$: contributes $\int_0^2 0 dx = 0$. Leg 2 ($x = 2$, $y: 0 \rightarrow 3$): $\mathbf{F} = \langle y, 2 \rangle$, $d\mathbf{r} = \langle 0, dy \rangle$: contributes $\int_0^3 2 dy = 6$. Total work $= 0 + 6 = 6$.
- 12 Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.
 $W = f(2, 3) - f(0, 0) = 6 - 0 = 6$, matching the piecewise calculation — confirming $\mathbf{F}$ is indeed conservative with potential $f(x, y) = xy$.

Testing for conservative fields

- 13 Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.
 $P = y^2$: $\frac{\partial P}{\partial y} = 2y$. $Q = 2xy + 1$: $\frac{\partial Q}{\partial x} = 2y$. Equal, so conservative. $f_x = y^2 \Rightarrow f = xy^2 + g(y)$;
 $f_y = 2xy + g'(y) = 2xy + 1$, so $g'(y) = 1$, $g(y) = y$. Potential function: $f(x, y) = xy^2 + y$.
- 14 Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.
 $P = 3x^2y$: $\frac{\partial P}{\partial y} = 3x^2$. $Q = x^3 + 2y$: $\frac{\partial Q}{\partial x} = 3x^2$. Equal, so conservative. $f_x = 3x^2y \Rightarrow f = x^3y + g(y)$;
 $f_y = x^3 + g'(y) = x^3 + 2y$, so $g'(y) = 2y$, $g(y) = y^2$. Potential function: $f(x, y) = x^3y + y^2$.

Applying Green's Theorem to circulation

- 15 Use Green's Theorem to evaluate the circulation $\oint_C (-y) dx + x dy$ around the boundary of the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$, traversed counterclockwise.
 $P = -y$, $Q = x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2$. $\iint_D 2 dA = 2(3)(2) = 12$.

Line integrals over piecewise paths

- 16** Evaluate $\int_C y \, dx + x \, dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.
 Leg 1 ($y = 0, x: 0 \rightarrow 1$): $y \, dx + x \, dy = 0 \, dx + x(0) = 0$: contributes 0. Leg 2 ($x = 1, y: 0 \rightarrow 1$): $y(0) + 1 \, dy$: contributes $\int_0^1 1 \, dy = 1$. Total = $0 + 1 = 1$.
- 17** Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.
 $f(1, 1) - f(0, 0) = 1 - 0 = 1$, matching the piecewise calculation.

Flux and Green's Theorem (normal form)

- 18** Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \iint_D \nabla \cdot \mathbf{F} \, dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \leq 4$.
 $\nabla \cdot \mathbf{F} = 1 + 1 = 2$. $\iint_D 2 \, dA = 2 \cdot (\pi \cdot 2^2) = 8\pi$.

A capstone conservative-field application

- 19** A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.
 Test: $P = 2xy^3: \frac{\partial P}{\partial y} = 6xy^2$. $Q = 3x^2y^2 + 4: \frac{\partial Q}{\partial x} = 6xy^2$. Equal, so conservative. Potential:
 $f_x = 2xy^3 \Rightarrow f = x^2y^3 + g(y)$; $f_y = 3x^2y^2 + g'(y) = 3x^2y^2 + 4$, so $g'(y) = 4$, $g(y) = 4y$:
 $f(x, y) = x^2y^3 + 4y$. Work = $f(2, 1) - f(0, 0) = (4 + 4) - 0 = 8$.
- 20** For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:
- a** Find the divergence $\nabla \cdot \mathbf{F}$. $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy + 2z) = 0 + 0 + 2 = 2$.
- b** Find the curl $\nabla \times \mathbf{F}$.
 $\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(xy + 2z) - \frac{\partial}{\partial z}(xz), \frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy + 2z), \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz) \right\rangle = \langle x - x, y - y, z - z \rangle = \langle 0, 0, 0 \rangle$.
- c** Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D. Since $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, $\mathbf{F}$ is conservative (irrotational) in 3D.